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# Censoring-Aware and Time-Varying Load Apportionment Modelling of River Water Quality

Comparing generalized additive, classical and hierarchical Bayesian models

by

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## Executive Summary

River managers need to know whether pollution mainly comes from fixed outlets or from wider areas of a catchment. This matters because the two sources usually need different actions. Point-source pollution may be reduced through treatment. Diffuse pollution often needs changes across the catchment. This project studied whether routine measurements of concentration and river flow can help separate these two types of behaviour. The analysis covered ammonia, nitrate, orthophosphate and dissolved zinc at six river sites in England.

Two features of the data were treated carefully. First, values below a detection limit were kept as censored observations. They were not replaced by the limit or by half of the limit. Second, repeated concentration samples taken on the same day were averaged before they were matched to daily mean flow. This stopped days with intensive sampling from having too much influence. All models were tested using the same chronological 80/20 training and test split.

A censoring-aware classical Load Apportionment Model (LAM) was first fitted to each site–determinand group. The model represents concentration as the sum of a point-source component, which decreases with flow, and a diffuse-source component, which increases with flow. The classical LAM was generally stable and performed particularly well for nitrate. Two censoring-aware generalized additive models (GAMs) were then fitted as flexible benchmarks. The first GAM included a long-term time trend and annual seasonality, while the second also included flow. Compared with the time-and-season GAM, adding flow improved test negative log-likelihood in 18 of the 23 groups. Randomised quantile residuals showed that the classical LAM was, on average, better calibrated than both GAM benchmarks.

A hierarchical Bayesian seasonal LAM was then fitted separately for each determinand. Site-level point and diffuse source sizes were partially pooled across sites. The flow exponents were shared within each determinand, and annual seasonality was modelled using one sine and one cosine term. A time-varying version then allowed the point and diffuse source sizes to change linearly through time.

The time-varying model was assessed using three test measures: negative log-likelihood, which evaluates the full predictive distribution and is lower when predictions are better; mean absolute error, which gives the average size of the prediction error; and root mean squared error, which gives more weight to larger errors. For ammonia, all three measures became worse, so the static model was preferred. For nitrate, negative log-likelihood improved only slightly, while the two error measures became slightly worse, which again suggested that the static model was sufficient. The clearest improvements were found for orthophosphate and zinc. For orthophosphate, mean test negative log-likelihood decreased from 0.693 to 0.293 and RMSE decreased from 0.352 to 0.282. Zinc showed the largest gain, with mean test negative log-likelihood decreasing from 0.200 to $-0.400$ and RMSE decreasing from 0.236 to 0.164.

The results showed clear differences between determinands. Nitrate was described well by a static concentration–flow relationship, with little evidence that time-varying source terms improved prediction. Orthophosphate benefited from the dynamic model, which showed a general decline in point-source contributions and a growing relative importance of diffuse sources at several sites. Zinc showed the largest predictive improvement, but its temporal patterns were more site-specific: some sites became relatively more diffuse dominated, while others became more point-source dominated. Ammonia remained the least reliable determinand because of heavy censoring. Overall, the study found that the appropriate model depends on the determinand, with static structure sufficient for nitrate and time-varying source contributions most useful for orthophosphate and dissolved zinc.

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## University of Edinburgh – Own Work Declaration

Name: Fakhruddin Hussain

I confirm that all this work is my own except where indicated.

Date: 26th June 2026

## Generative AI Disclosure

Generative AI was used during this project and while preparing the report.

- **Report:** ChatGPT was used to help plan the report and for correcting grammatical mistakes.
- **Mathematics:** ChatGPT was used to brain storm ideas and for figuring out some mathematical concepts like censoring-aware likelihood.

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# 1 Introduction

The concentration of a substance in a river depends on both the amount entering the river and the amount of water available to carry it. A fairly steady discharge from a fixed outlet can produce a concentration that falls as flow rises. By contrast, rainfall can wash substances from soils, drains and other parts of a catchment into the river, so concentration may rise during high-flow periods. Concentration–flow relationships can therefore give useful evidence about pollution sources.

The load allocation model (LAM) uses this idea. It writes concentration as the sum of a point-source term and a diffuse-source term, both expressed as power functions of flow (Bowes, Smith, et al., 2008). The method has been used to help identify priorities for nutrient control in river basins, including the Thames (Bowes, Jarvie, et al., 2014). Its main advantage is that it only needs paired concentration and flow observations. However, the usual LAM assumes that its parameters stay fixed over the full monitoring period.

This project deals with three main statistical problems. First, environmental data often contain values below a detection limit. Replacing these values by the limit or by half of the limit treats an uncertain value as exact and can bias the analysis (**helsel2012**). Second, a fixed LAM may be too simple when treatment systems, land use, laboratory methods or river behaviour change over time. Third, separate models for each site can give unstable source estimates when some sites have few observations or when the two LAM components have similar shapes.

Generalized additive models (GAMs) provide a useful comparison because they can estimate smooth nonlinear relationships without forcing a point-source and diffuse-source split (**wood2017**). Comparing GAMs and LAMs on the same test data shows the balance between flexible prediction and source interpretation. A Bayesian hierarchical model can also share information across sites while still allowing sites to differ. It also gives posterior intervals for the source shares.

The main aim was to find out when a static or time-varying LAM gives a useful description of daily river-water concentration. The study had five objectives:

1. build a daily dataset that keeps exact, left-censored and interval-censored information;
2. test whether flow adds useful information beyond time trend and annual seasonality;
3. compare GAMs, a classical static LAM and hierarchical Bayesian LAMs using the same chronological test split;
4. check whether the point and diffuse components are well enough identified to interpret; and
5. find which determinands benefit from time-varying source sizes.

# 2 Data and preprocessing

## 2.1 Monitoring data

Concentration data came from the Environment Agency Water Quality Archive, and daily mean flow came from the Environment Agency Hydrology Data Explorer (Environment Agency, 2026b; Environment Agency, 2026a). Six sites were included: Thames at Teddington, Severn at Haw Bridge, Tees at Low Worsall, Eden at Sheepmount, Trent at Dunham and Wensum at Sweet Briar Road Bridge. Four determinands were studied: ammonia, nitrate, orthophosphate and dissolved zinc.

The raw concentration data contained 16,233 valid records. There were no missing dates or results, no non-positive values, no unexpected qualifiers and no exact duplicate records. However, the amount of data differed a lot between sites and years. Eden and Wensum had many observations, and sampling frequency changed over time. Eden had only one dissolved-zinc observation, which was censored, so this group was removed. This left 23 usable site–determinand groups.

The percentage of raw censored observations was about 42.6% for ammonia, 1.9% for nitrate, 9.5% for orthophosphate and 17.5% for zinc. These percentages varied strongly by site. Most detection limits were stable, but there were a few unusual values in the Tees data. Wensum zinc also showed a clear change in reporting after 2015, when values below 5 were more often reported exactly.

**Table 1:** Study design and its effect on the modelling approach.

| Item | Included categories | Effect on modelling |
|---|---|---|
| **Sites** | Thames, Severn, Tees, Eden, Trent and Wensum | Site differences were kept in the models. |
| **Determinand** | Ammonia, nitrate, orthophosphate and dissolved zinc | Bayesian models were fitted separately for each determinand. |
| **Response status** | Exact, left-censored and interval-censored daily concentration | The likelihood used the observed bounds instead of substituted values. |
| **Predictor** | Daily mean river flow | Concentration and flow were matched by site and date. |
| **Validation** | Earliest 80% for training and latest 20% for testing | All models were tested on the same future time block. |

## 2.2 Daily aggregation and censoring

There were 914 site–determinand–date groups with more than one concentration sample. These groups contained 3,920 records, with up to 11 samples on one day. If each sample had been matched to the same daily flow, heavily sampled days would have had too much weight. The concentration data were therefore reduced to one record for each site, determinand and date before taking logarithms.

For a day with $n$ measurements, let $E$ be the exact measurements and $L$ the left-censored measurements. Let $u_j$ be the reported upper limit for censored value $j$. The daily response was defined as

$$\text{all exact:} \quad C_{\text{day}} = \frac{1}{n} \sum_{i \in E} y_i, \tag{1}$$

$$\text{all censored:} \quad 0 < C_{\text{day}} \leq \frac{1}{n} \sum_{j \in L} u_j, \tag{2}$$

$$\text{mixed:} \quad \frac{1}{n} \sum_{i \in E} y_i < C_{\text{day}} \leq \frac{1}{n} \left( \sum_{i \in E} y_i + \sum_{j \in L} u_j \right). \tag{3}$$

All-exact days stayed exact. All-censored days stayed left-censored. Days with both exact and censored samples became interval-censored. The final daily dataset contained 13,227 records: 10,816 exact days, 2,280 all-censored days and 131 mixed days.

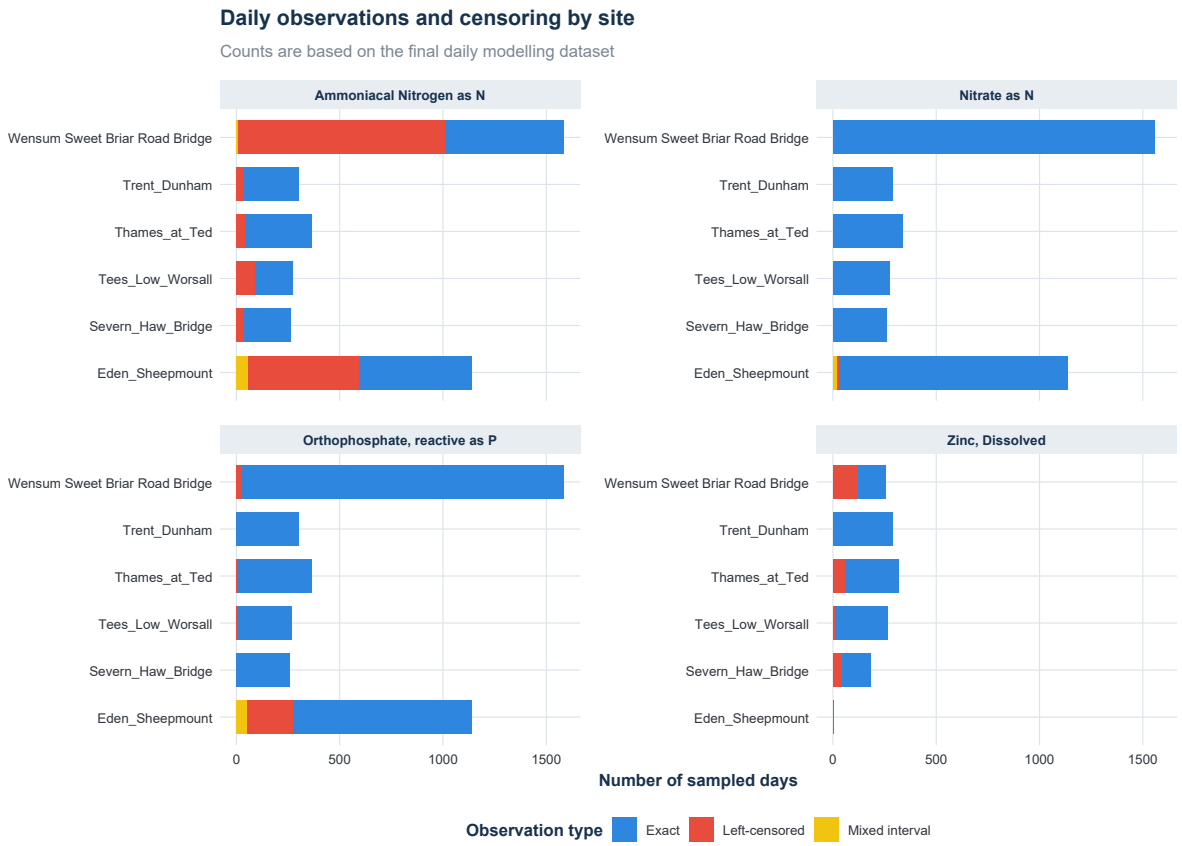



**Figure 1:** Coverage, sampling frequency and censoring in the daily modelling data.

### 2.3 Training and test data

The observations in each site–determinand group were ordered by date. The earliest 80% were used for training, and the latest 20% were used for testing. A chronological split was chosen to keep consistency. Every model used the same saved split, so the comparisons were fair.

## 3 Methods

### 3.1 Censoring-aware observation likelihood

Let $C_{is} > 0$ be the true concentration for observation $i$ at site $s$. The response was modelled on the $\log_{10}$ scale:

$$Y^*_{is} = \log_{10} C_{is}. \tag{4}$$

The observation model was

$$Y^*_{is} \sim \mathcal{N}\left(\mu_{is}, \sigma^2_{is}\right), \tag{5}$$

where $\mu_{is}$ is the mean log concentration and $\sigma_{is} > 0$ is the residual standard deviation.

Three types of daily observations were present. For an exact observation, $y_{is}$ is the observed log concentration. For a left-censored observation, $u_{is}$ is the log of the upper detection limit. For an interval-censored observation, $l_{is}$ and $u_{is}$ are the logs of the lower and upper bounds.

Let $\phi$ and $\Phi$ denote the standard Normal density and distribution functions. The likelihood contribution was

$$L_{is} = \begin{cases} \frac{1}{\sigma_{is}} \phi\left(\frac{y_{is} - \mu_{is}}{\sigma_{is}}\right), & \text{exact,} \\ \Phi\left(\frac{u_{is} - \mu_{is}}{\sigma_{is}}\right), & \text{left-censored,} \\ \Phi\left(\frac{u_{is} - \mu_{is}}{\sigma_{is}}\right) - \Phi\left(\frac{l_{is} - \mu_{is}}{\sigma_{is}}\right), & \text{interval-censored.} \end{cases} \tag{6}$$

Therefore, an exact observation contributed through its density, while a censored observation contributed through the probability of lying inside its known range. Censored observations were not replaced by assumed exact concentrations.

## 3.2 Generalized additive benchmarks

Two generalized additive models were fitted separately to each site–determinand group. Both used a censored-Normal response model in `mgcv`.

The first GAM includes long-term change and annual seasonality:

$$\mu_i = \beta_0 + f_t(t_i) + f_d(d_i), \tag{7}$$

where:

- $\beta_0$ is the intercept;
- $t_i$ is continuous calendar time;
- $d_i$ is day of year;
- $f_t(\cdot)$ is a smooth long-term time effect; and
- $f_d(\cdot)$ is a cyclic annual smooth.

The second GAM also included river flow:

$$\mu_i = \beta_0 + f_t(t_i) + f_d(d_i) + f_Q\left(\log_{10} Q_i\right), \tag{8}$$

where $Q_i > 0$ is daily mean flow and $f_Q(\cdot)$ is a smooth flow effect. Smoothness was selected using restricted maximum likelihood.

These models provided flexible prediction benchmarks. However, second GAM does not separate concentration into point-source and diffuse-source components.

## 3.3 Classical static Load Apportionment Model

The classical Load Apportionment Model was fitted separately to each site–determinand group. For daily mean flow $Q_i > 0$, the model was

$$C_i = AQ_i^{B-1} + MQ_i^{D-1}. \tag{9}$$

The four model parameters have the following interpretation:

- $A > 0$ controls the size of the point-source component;
- $M > 0$ controls the size of the diffuse-source component;
- $0 < B < 1$ controls how the point-source component changes with flow; and
- $D > 1$ controls how the diffuse-source component changes with flow.

The point-source component is

$$C_{P,i} = AQ_i^{B-1}. \tag{10}$$

Because $B - 1 < 0$, this component decreases as flow increases.

The diffuse-source component is

$$C_{D,i} = MQ_i^{D-1}. \tag{11}$$

Because $D - 1 > 0$, this component increases with flow. This represents material being washed into the river from the surrounding catchment during wetter conditions. (Bowes, Smith, et al., 2008).

The mean of the modelled log concentration scale was

$$\mu_i = \log_{10}\left(AQ_i^{B-1} + MQ_i^{D-1}\right). \tag{12}$$

The parameter restrictions were imposed using unconstrained parameters $a$, $m$, $b$ and $d$:

$$A = \exp(a), \qquad M = \exp(m), \qquad B = \text{logit}^{-1}(b), \qquad D = 1 + \exp(d). \tag{13}$$

These transformations ensure that $A$ and $M$ are positive, $0 < B < 1$, and $D > 1$.

The model was estimated by maximising the censoring-aware likelihood in equation (6). Several fixed starting values were used because nonlinear optimisation can reach different local solutions. Numerical convergence was checked separately.

At any flow $Q$, the point-source share was

$$s_P(Q) = \frac{AQ^{B-1}}{AQ^{B-1} + MQ^{D-1}}. \tag{14}$$

The diffuse-source share was

$$s_D(Q) = 1 - s_P(Q). \tag{15}$$

These shares estimate how much of the concentration comes from each source at a given flow.

If $B \neq D$, the flow at which the two components were equal was

$$Q_\times = \left(\frac{A}{M}\right)^{1/(D-B)}. \tag{16}$$

## 3.4 Hierarchical Bayesian static seasonal LAM

A hierarchical Bayesian model was fitted for each determinand. Sites were fitted together within each determinand, allowing information to be shared across sites.

Flow was first divided by a determinand-specific reference flow:

$$q_{is} = \frac{Q_{is}}{Q_{\text{ref}}}, \tag{17}$$

where $Q_{is}$ is the observed flow and $Q_{\text{ref}} > 0$ is the reference flow. Therefore, $q_{is} = 1$ at the reference flow.

Two site-specific linear predictors were used:

$$a_s = \alpha_A + u_{A,s}, \tag{18}$$

$$m_s = \alpha_M + u_{M,s}. \tag{19}$$

Here:

- $\alpha_A$ is the average point-source level across sites;
- $\alpha_M$ is the average diffuse-source level across sites;
- $u_{A,s}$ is the point-source deviation for site $s$; and
- $u_{M,s}$ is the diffuse-source deviation for site $s$.

The positive source coefficients were then

$$A_s = \exp(a_s), \qquad M_s = \exp(m_s). \tag{20}$$

The flow exponents were

$$B = \text{logit}^{-1}(b), \qquad D = 1 + \exp(d). \tag{21}$$

The parameters $b$ and $d$ are unconstrained, while the transformations ensure $0 < B < 1$ and $D > 1$.

The LAM concentration at site $s$ was

$$C_{is}^{\mathrm{LAM}} = A_s q_{is}^{B-1} + M_s q_{is}^{D-1}. \tag{22}$$

A separate annual seasonal effect was added on the $\log_{10}$ scale:

$$h(d_{is}) = \gamma_{\sin} \sin\left(\frac{2\pi d_{is}}{365.25}\right) + \gamma_{\cos} \cos\left(\frac{2\pi d_{is}}{365.25}\right), \tag{23}$$

where $d_{is}$ is day of year. The two coefficients $\gamma_{\sin}$ and $\gamma_{\cos}$ control the size and timing of the annual cycle.

The final mean was

$$\mu_{is} = \log_{10}\left(C_{is}^{\mathrm{LAM}}\right) + h(d_{is}). \tag{24}$$

The flow exponents and seasonal coefficients were shared across sites within a determinand. The source coefficients $A_s$ and $M_s$ were allowed to differ by site through partial pooling.

## 3.5 Hierarchical Bayesian time-varying LAM

The time-varying model used the same source structure, flow effects and seasonal pattern as the static Bayesian model. The only difference was that the point-source and diffuse-source sizes could increase or decrease steadily over time.

Let $z_{is}$ be centred and scaled calendar time. A value of $z_{is} = 0$ represents the centre of the observed time period. Negative values represent earlier observations and positive values represent later observations.

The point-source linear predictor was

$$a_s(z_{is}) = \alpha_A + u_{A0,s} + \left(\beta_A + u_{A1,s}\right) z_{is}. \tag{25}$$

The diffuse-source linear predictor was

$$m_s(z_{is}) = \alpha_M + u_{M0,s} + \left(\beta_M + u_{M1,s}\right) z_{is}. \tag{26}$$

In these equations:

- $\alpha_A$ and $\alpha_M$ are the average source levels at the centre of the study period;
- $u_{A0,s}$ and $u_{M0,s}$ are site-specific differences in those source levels;
- $\beta_A$ and $\beta_M$ are the average time slopes across sites; and
- $u_{A1,s}$ and $u_{M1,s}$ are site-specific differences in the time slopes.

The positive time-varying source coefficients were

$$A_s(z_{is}) = \exp(a_s(z_{is})), \qquad M_s(z_{is}) = \exp(m_s(z_{is})). \tag{27}$$

For example, a positive value of $\beta_A + u_{A1,s}$ indicates that the point-source coefficient increased through time at site $s$. A value close to zero indicates little systematic temporal change.

The time-varying LAM concentration was

$$C_{is}^{\mathrm{LAM}} = A_s(z_{is}) q_{is}^{B-1} + M_s(z_{is}) q_{is}^{D-1}. \tag{28}$$

The mean log concentration was

$$\mu_{is} = \log_{10}\left(C_{is}^{\mathrm{LAM}}\right) + h(d_{is}), \tag{29}$$

where $h(d_{is})$ is the same annual sine–cosine effect as in equation (23).

The exponents $B$ and $D$ remained constant through time within each determinand. Only the source coefficients $A_s(z)$ and $M_s(z)$ were allowed to vary through time.

For posterior draw $r$, the time-varying point-source share was

$$s_{P,is}^{(r)} = \frac{A_s^{(r)}(z_{is}) q_{is}^{B^{(r)}-1}}{A_s^{(r)}(z_{is}) q_{is}^{B^{(r)}-1} + M_s^{(r)}(z_{is}) q_{is}^{D^{(r)}-1}}. \tag{30}$$

The diffuse-source share was

$$s_{D,is}^{(r)} = 1 - s_{P,is}^{(r)}. \tag{31}$$

Posterior medians and 95% credible intervals were used to summarise the time-varying source coefficients and source shares.

## 3.6 Prior specification and sensitivity analysis

Simple priors were used to keep the estimates stable without forcing the model towards a particular source split. Before fitting the model, simulated data were generated from the priors to check that they produced realistic concentration values.

The exact prior distributions used in the fitted models are shown in table 2. In the table, $\mathcal{N}^{+}(0, s^2)$ denotes a Normal distribution with mean zero and standard deviation $s$, truncated below at zero. The notation $t_3^{+}(0, s)$ denotes a Student-$t$ distribution with three degrees of freedom, location zero and scale $s$, also truncated below at zero.

**Table 2:** Exact prior distributions used in the hierarchical static and time-varying Bayesian LAMs. The second parameter of each Normal distribution is its standard deviation.

| Parameter or effect | Static Bayesian LAM | Time-varying Bayesian LAM |
|---|---|---|
| Average source levels | $\alpha_A, \alpha_M \sim \mathcal{N}(-0.3, 0.7^2)$ | $\alpha_A, \alpha_M \sim \mathcal{N}(-0.3, 0.7^2)$ |
| Site differences in source levels | $u_{A,s} \mid \tau_A \sim \mathcal{N}(0, \tau_A^2)$<br>$u_{M,s} \mid \tau_M \sim \mathcal{N}(0, \tau_M^2)$<br>$\tau_A, \tau_M \sim \mathcal{N}^{+}(0, 0.5^2)$ | $u_{A0,s} \mid \tau_{A0} \sim \mathcal{N}(0, \tau_{A0}^2)$<br>$u_{M0,s} \mid \tau_{M0} \sim \mathcal{N}(0, \tau_{M0}^2)$<br>$\tau_{A0}, \tau_{M0} \sim \mathcal{N}^{+}(0, 0.4^2)$ |
| Average time slopes | Not included | $\beta_A, \beta_M \sim \mathcal{N}(0, 0.3^2)$ |
| Site differences in time slopes | Not included | $u_{A1,s} \mid \tau_{A1} \sim \mathcal{N}(0, \tau_{A1}^2)$<br>$u_{M1,s} \mid \tau_{M1} \sim \mathcal{N}(0, \tau_{M1}^2)$<br>$\tau_{A1}, \tau_{M1} \sim \mathcal{N}^{+}(0, 0.4^2)$ |
| Point-source flow parameter | $b \sim \mathcal{N}(0, 1^2)$ | $b \sim \mathcal{N}(0, 1^2)$ |
| Diffuse-source flow parameter | $d \sim \mathcal{N}(-0.7, 0.7^2)$ | $d \sim \mathcal{N}(-0.7, 0.7^2)$ |
| Seasonal coefficients | $\gamma_{\sin}, \gamma_{\cos} \sim \mathcal{N}(0, 0.2^2)$ | $\gamma_{\sin}, \gamma_{\cos} \sim \mathcal{N}(0, 0.2^2)$ |
| Residual variation | $\sigma \sim t_3^{+}(0, 0.3)$ | Every residual-scale coefficient had $\kappa_j \sim \mathcal{N}(-1.2, 0.8^2)$ |

A focused prior-sensitivity analysis was performed for the accepted time-varying orthophosphate model. The prior means were unchanged. The prior standard deviations were multiplied by 0.67 for the tighter specification and by 1.5 for the wider specification.

**Table 3:** Results of the focused prior-sensitivity analysis for the time-varying orthophosphate model. Prediction differences are measured against the baseline model on the $\log_{10}$ concentration scale.

| Measure | Baseline | Tight | Wide |
|---|---|---|---|
| Maximum $\widehat{R}$ | 1.010 | 1.025 | 1.004 |
| Minimum bulk ESS | 454 | 137 | 725 |
| Minimum tail ESS | 458 | 763 | 1055 |
| Divergent transitions | 0 | 0 | 0 |
| Maximum tree depth | 11 | 11 | 12 |
| Posterior median of $B$ | 0.487 | 0.461 | 0.493 |
| Posterior median of $D$ | 1.927 | 1.842 | 1.956 |
| Median between-site SD of the point-source time slope | 1.273 | 0.489 | 1.562 |
| Mean absolute prediction difference | – | 0.043 | 0.010 |
| Prediction correlation | – | 0.987 | 0.9996 |
| Maximum absolute prediction difference | – | 1.152 | 0.182 |
| Predictions differing by more than 0.1 | – | 12.5% | 1.0% |

The posterior medians of $B$ and $D$ changed only slightly across the three prior settings. The annual seasonal coefficients were also very similar. The flow response and seasonal pattern were therefore not strongly controlled by the baseline prior choices.

The main sensitivity was in the amount of between-site variation in the point-source time slopes. The tighter priors reduced its posterior median from 1.273 to 0.489, while the wider priors increased it to 1.562. Therefore, stronger priors produced more shrinkage of the site-specific temporal trends.

Predictions under the wider priors were almost identical to the baseline predictions. Their correlation was 0.9996 and their mean absolute difference was 0.010. The tighter-prior predictions were also strongly correlated with the baseline predictions, although a small group of predictions changed more noticeably.

The tighter fit had weaker sampling behaviour, with a maximum $\widehat{R}$ of 1.025 and a minimum bulk effective sample size of 137. Overall, the flow response, annual seasonality and most predictions were robust, but the estimated size of site-specific temporal variation was more sensitive to the prior choice.

## 3.7 Evaluation and diagnostics

The main held-out performance measure was mean negative log-likelihood:

$$\mathrm{NLL} = -\frac{1}{n_{\mathrm{test}}} \sum_{i \in \mathrm{test}} \log L_i, \tag{32}$$

where $L_i$ is the likelihood contribution in equation (6). Lower NLL indicates better predictive performance.

For the set $E$ of exact test observations, three further measures were calculated:

$$\mathrm{MAE} = \frac{1}{n_E} \sum_{i \in E} |y_i - \widehat{\mu}_i|, \tag{33}$$

$$\mathrm{RMSE} = \left\{ \frac{1}{n_E} \sum_{i \in E} (y_i - \widehat{\mu}_i)^2 \right\}^{1/2}, \tag{34}$$

$$\mathrm{Bias} = \frac{1}{n_E} \sum_{i \in E} (\widehat{\mu}_i - y_i). \tag{35}$$

MAE and RMSE measure the size of prediction errors on the $\log_{10}$ concentration scale. Bias shows whether predictions were too high or too low on average.

Randomised quantile residuals were used to assess exact and censored observations on the same scale (Dunn and Smyth,

1996). Let $F_i$ denote the fitted distribution function for observation $i$. A probability-scale value $U_i$ was calculated as

$$U_i = \begin{cases} F_i(y_i), & \text{exact,} \\ \text{Uniform}(0, F_i(u_i)), & \text{left-censored,} \\ \text{Uniform}(F_i(l_i), F_i(u_i)), & \text{interval-censored.} \end{cases} \tag{36}$$

The residual was then

$$r_i = \Phi^{-1}(U_i). \tag{37}$$

For a well-calibrated model, these residuals should have a mean close to zero, a standard deviation close to one, and little remaining relationship with time, flow or the previous residual.

Bayesian sampling was assessed using rank-normalised $\widehat{R}$, bulk and tail effective sample sizes, divergent transitions, maximum tree depth and posterior predictive checks (Bürkner, 2017; Vehtari et al., 2021). A more complex model was retained only when its diagnostics were acceptable and it provided a meaningful improvement in held-out prediction.

# 4 Results

## 4.1 Exploratory concentration–flow patterns

The relationship between concentration and flow differed by both site and determinand. Orthophosphate was the closest to the expected LAM shape overall, but the direction still varied. Concentration increased with flow at Eden, mostly decreased at Severn, Tees, Thames and Trent, and showed a weak U-shape at Wensum. Nitrate usually decreased with flow, while Wensum showed a pattern that does not consistently increase or decrease. Ammonia had heavy censoring and a large change in the Eden series. Zinc showed the weakest and most site-specific relationship, and the Wensum data were affected by a likely change in reporting sensitivity.

These results show that one concentration–flow curve cannot describe every site. They also show why plots based only on exact observations must be treated with care. Removing censored observations removes many low concentrations. Figure 2 should therefore show the censoring status directly.




**Figure 2:** Daily log concentration against log daily mean flow for each site and determinand.

## 4.2 Classical LAM and GAM benchmarks

The classical LAM was fitted first to each site–determinand group. It achieved a site-balanced mean test NLL of −0.0589, a mean absolute error (MAE) of 0.172 and a root mean squared error (RMSE) of 0.213 on the $\log_{10}$ concentration scale. Its strongest results were for nitrate.

Two GAMs were then fitted as flexible benchmarks. The first included time and seasonality, while the second also included flow. Adding flow improved mean test NLL in 18 of the 23 groups. This included all six ammonia groups, five of the six nitrate groups, five of the six orthophosphate groups and two of the five zinc groups. The main cases where adding flow made prediction worse were Wensum nitrate, Wensum orthophosphate and Tees zinc. This suggests that the concentration–flow relationship learned from the training period did not continue in the same way during the test period at these sites.

The classical LAM was then compared with the GAM that included time, seasonality and flow. The LAM had lower NLL in 12 of the 23 groups, lower MAE in 14 groups and lower RMSE in 15 groups. It performed especially well for nitrate, where it was better than the full GAM on all three measures at five of the six sites. The results for orthophosphate and zinc were more mixed. For ammonia, the likelihood and error measures did not always agree because a large proportion of the observations were censored.

**Table 4:** Test performance of the classical LAM and the two GAM benchmarks, averaged across sites within each determinand. Lower values are better.

| Determinand | Model | Mean NLL | MAE ($\log_{10}$) | RMSE ($\log_{10}$) |
|---|---|---|---|---|
| Ammonia | Classical LAM | 0.465 | 0.236 | 0.298 |
| | Time–season GAM | 0.364 | 0.273 | 0.335 |
| | Time–season–flow GAM | 0.346 | 0.308 | 0.371 |
| Nitrate | Classical LAM | -0.871 | 0.082 | 0.104 |
| | Time–season GAM | 4.760 | 0.237 | 0.267 |
| | Time–season–flow GAM | 2.245 | 0.126 | 0.150 |
| Orthophosphate | Classical LAM | 0.313 | 0.212 | 0.252 |
| | Time–season GAM | 0.305 | 0.192 | 0.235 |
| | Time–season–flow GAM | 0.405 | 0.161 | 0.200 |
| Dissolved zinc | Classical LAM | -0.160 | 0.158 | 0.192 |
| | Time–season GAM | 0.053 | 0.170 | 0.205 |
| | Time–season–flow GAM | -0.279 | 0.142 | 0.180 |

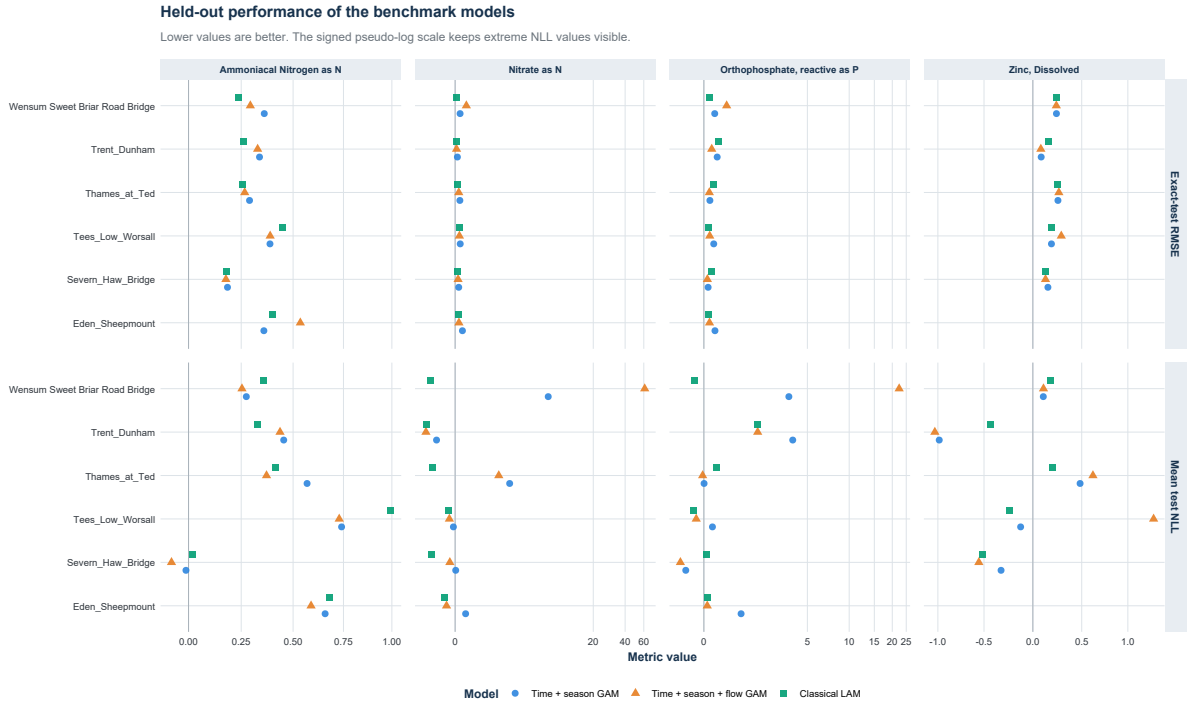



**Figure 3:** Test mean NLL and exact-test RMSE for the time–season GAM, time–season–flow GAM and classical static LAM.

## 4.3 Calibration, identifiability and source components

On the test data, the classical LAM had better average RQR results than the two GAMs. The mean absolute residual mean was 0.461 for the LAM, compared with 0.949 for the time–season GAM and 1.02 for the time–season–flow GAM. The mean absolute difference between residual standard deviation and one was 0.171 for the LAM, compared with 0.293 and 0.286 for the two GAMs. The mean percentage of residuals outside ±1.96 was 7.63% for the LAM and 21.4% for both GAMs. Even so, some series still had strong lag-one dependence, including Wensum nitrate, Wensum orthophosphate, Eden nitrate and Wensum zinc.

All 23 classical LAM fits returned optimisation convergence code zero. However, many parameter estimates were close to their limits, or one component was very small. Several estimates of $B$ were close to zero or one, and many estimates of $D$ were close to one. The component checks showed clear point-source dominance for several nitrate series, clear diffuse-source dominance for Eden orthophosphate and some Tees, Trent and Wensum series, and mixed behaviour in other groups. Twelve of the 23 crossover flows were inside the observed training-flow range.

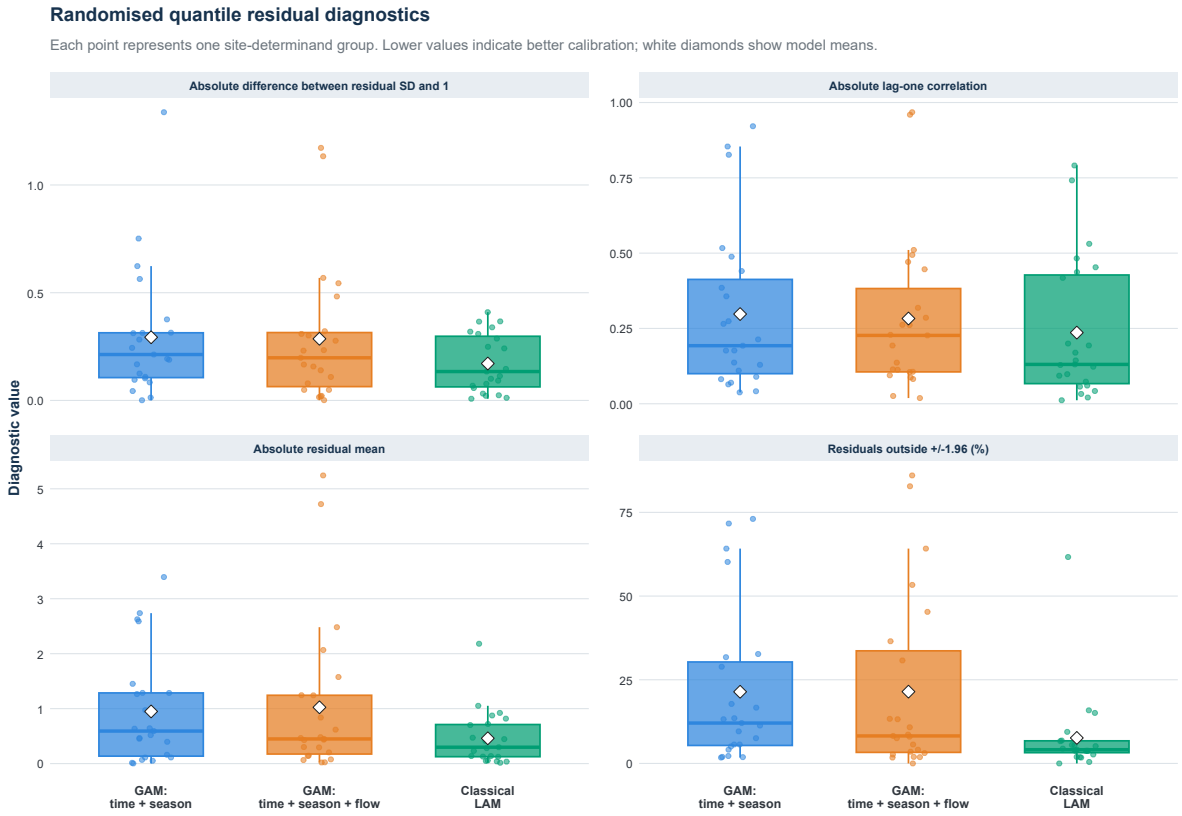



**Figure 4:** Randomised quantile residual checks for each site–determinand group and model.

## 4.4 Hierarchical Bayesian static and time-varying results

Table 5 compares the hierarchical static seasonal model with the linear time-varying model. The values are averages across sites within each determinand and use the same chronological test observations.

**Table 5:** Test performance of the hierarchical Bayesian static seasonal LAM and the linear time-varying LAM. Lower values are better.

| Determinand | Model | Mean NLL | MAE ($\log_{10}$) | RMSE ($\log_{10}$) |
|---|---|---|---|---|
| Ammonia | Static Bayesian | 1.305 | 0.920 | 1.031 |
| | Time-varying Bayesian | 1.362 | 1.288 | 1.422 |
| Nitrate | Static Bayesian | -0.836 | 0.077 | 0.101 |
| | Time-varying Bayesian | -0.845 | 0.083 | 0.103 |
| Orthophosphate | Static Bayesian | 0.693 | 0.314 | 0.352 |
| | Time-varying Bayesian | 0.293 | 0.244 | 0.282 |
| Dissolved zinc | Static Bayesian | 0.200 | 0.208 | 0.236 |
| | Time-varying Bayesian | -0.400 | 0.126 | 0.164 |

The time-varying model was not suitable for ammonia. NLL increased from 1.305 to 1.362, and RMSE increased from 1.031 to 1.422. A simple linear time trend could not describe the heavy censoring, the large differences between sites and the sudden high-concentration period at Eden.

For nitrate, the dynamic model reduced NLL by only 0.009, while MAE and RMSE increased slightly. The static relationship already predicted well, especially in the classical LAM. The static model is therefore preferred for nitrate.

Orthophosphate improved under the time-varying model. Mean NLL fell by 0.401, and RMSE fell by 0.071. This gives evidence that the point and/or diffuse source sizes changed over time, although the amount of change differed between sites.

Zinc had the largest improvement. Mean NLL fell by 0.600, MAE fell by 0.082 and RMSE fell by 0.072. This matches the visible change over time in the Wensum record and the remaining dependence under the static model.

**Key finding**

The time-varying model was not better for every determinand. The preferred choice is static for nitrate, dynamic for zinc, dynamic at selected sites for orthophosphate, and not reliable for ammonia without a different time structure.

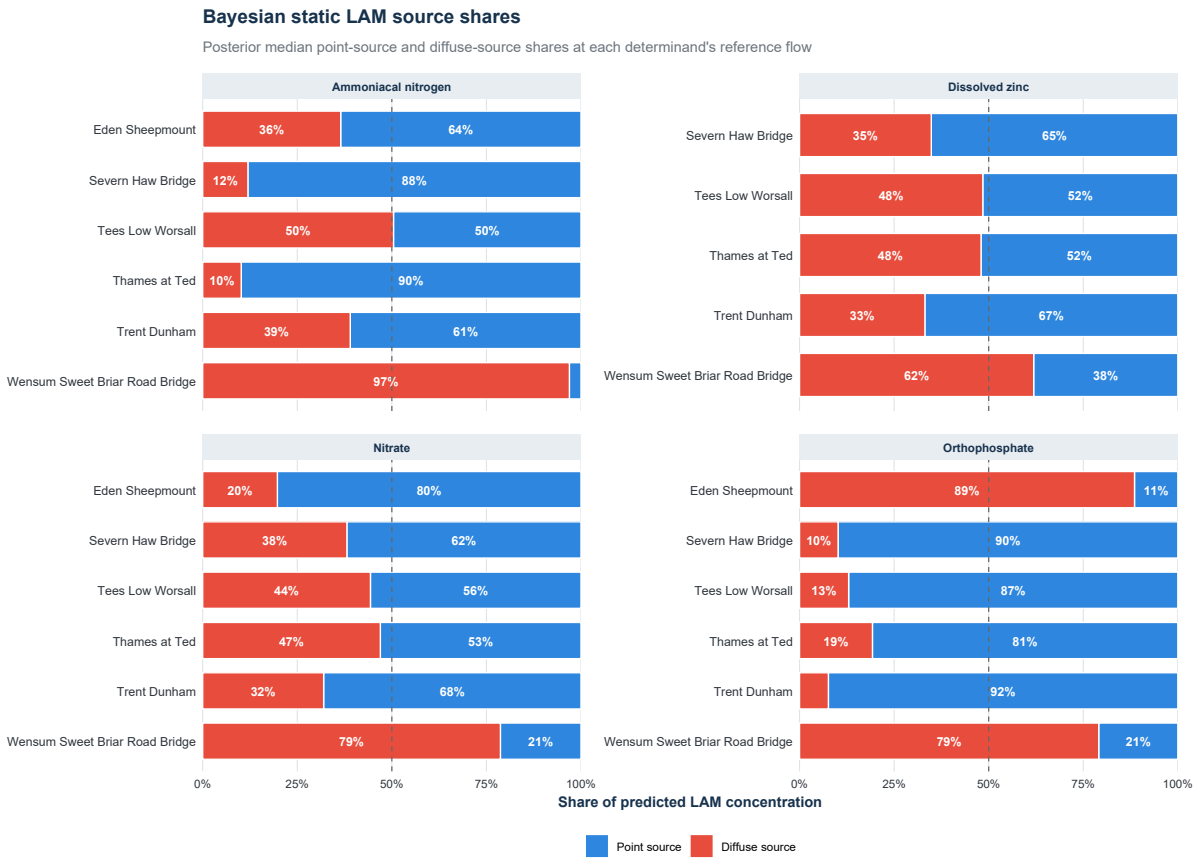



**Figure 5:** Posterior medians and 95% credible intervals for the point-source share from the static model at a common reference flow.

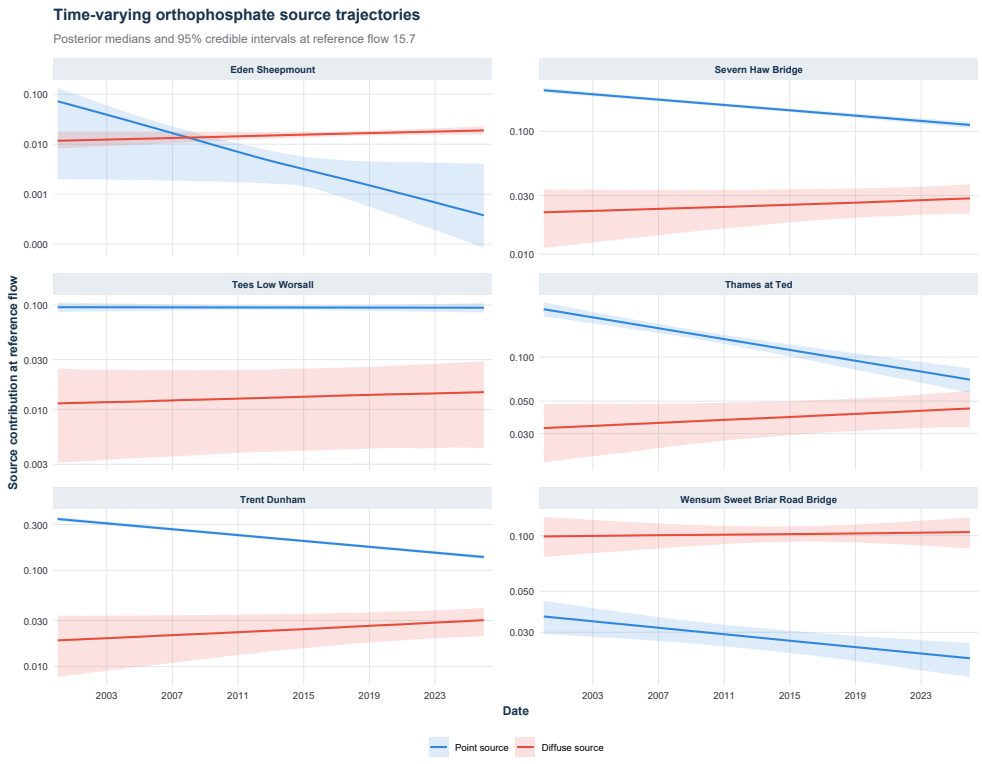



**Figure 6:** Posterior paths of the point-source and diffuse-source components for orthophosphate.

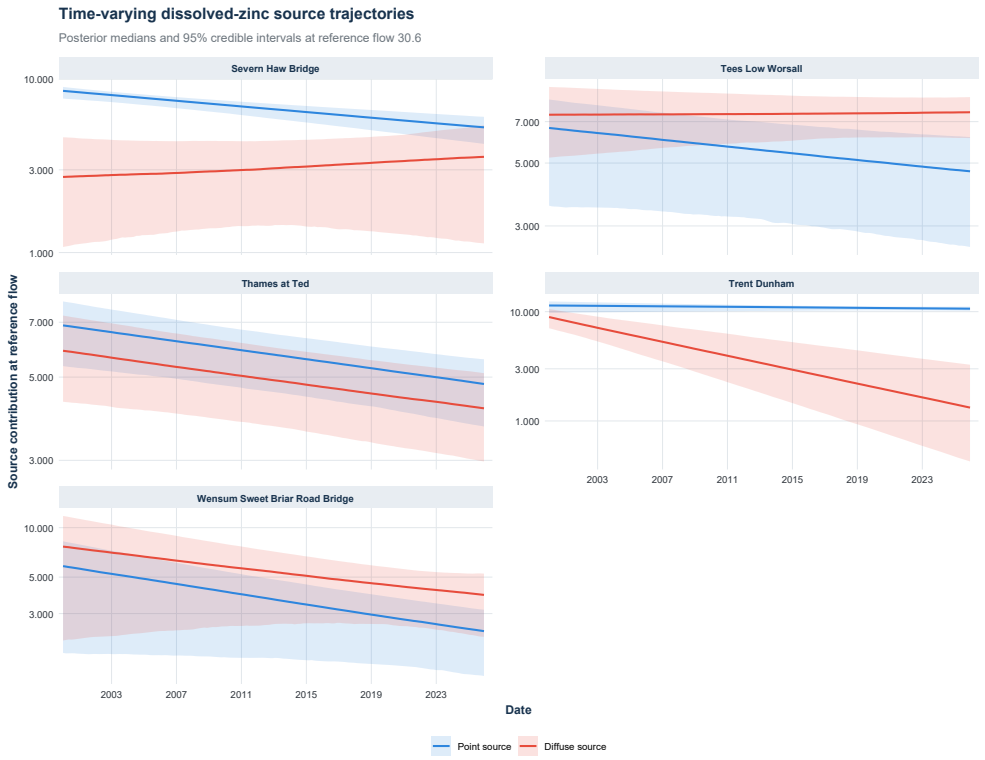



**Figure 7:** Posterior paths of the point-source and diffuse-source components for dissolved zinc from the selected full-data dynamic refit.

### 4.5 Posterior diagnostics and interpretation

The Bayesian comparisons used models with acceptable chain mixing for the reported prediction results. The checks included $\widehat{R}$, effective sample sizes and divergent transitions, as well as trace plots where needed.

The posterior source shares are not direct measurements of sewage, farming or any other physical source. They are the shares of the two model components in equation (30). Wide intervals, strong dependence between parameters, or shares close to zero or one can mean that total concentration is better identified than the point/diffuse split. This is especially important where the classical estimates were close to parameter limits.

## 5 Discussion

### 5.1 Prediction and scientific meaning

The first main result is that flow often helps prediction. Adding flow improved the GAM likelihood for most groups, especially nitrate and orthophosphate. However, the poor Wensum results show that a flexible smooth fitted to early data can fail when the relationship changes. The classical LAM was more stable because its shape is more restricted.

The second main finding is that the static LAM is still a useful baseline model. It predicted nitrate well and its residuals were, on average, better calibrated than those from the GAMs. However, good prediction of total concentration did not always mean that the point-source and diffuse-source parts were clearly identified. Different combinations of the two components could produce a similar overall concentration curve. For this reason, source contributions should be interpreted using their posterior uncertainty and by checking whether both components are supported by the observed flow data.

The third result is that time change depends on the determinand. The linear dynamic model gave a large gain for zinc and a smaller gain for orthophosphate. It gave almost no gain for nitrate and made ammonia worse. A dynamic model should not be chosen only because the data cover many years. It should be supported by better test prediction, acceptable posterior checks and a sensible scientific explanation.

### 5.2 Role of partial pooling and seasonality

A model with complete pooling would force all sites to have the same source sizes. Separate models would allow every site to differ, but weakly informed sites could then have unstable estimates. Partial pooling gives a middle option. Sites can have different estimates, but uncertain ones are moved closer to the average for that determinand. Using the same $B$ and $D$ for all sites within a determinand makes the model simpler. This means the time-varying part only tracks changes in source size, while the effect of flow stays fixed.

The annual sine and cosine terms give a simple seasonal pattern. They are easier to explain and fit than a more flexible cyclic spline. They separate a repeating yearly effect from a long-term rise or fall.

### 5.3 Limitations

Daily mean flow may not exactly match a concentration sample collected at a specific time of day. Using one value per day prevents heavily sampled days from having too much influence, but it also removes changes within the day. Where accurate sample times are available, matching them with higher-frequency flow data would give a better comparison.

Changes in laboratory reporting are another limitation. Before 2015, many Wensum zinc values below 5 were censored. Later, values below 5 were often reported exactly. This strongly suggests improved measurement sensitivity. The time-varying model can fit this shift, but it cannot tell whether the change came from the laboratory or the river.

The linear time model is deliberately simple. It can describe a steady increase or decrease, but not a sudden jump, reversal or curved path. This is one reason it performed poorly for ammonia. A change-point model or a more flexible time function may fit Eden ammonia better. These models were not added because the aim of this report was a clear and reproducible comparison, not a search through many complex models.

Finally, the point and diffuse labels come from the shapes of the two flow terms. Real point-source loads can change over time, diffuse concentration does not always rise with flow, and material stored in the river can affect both patterns.

### 5.4 Practical recommendations

For nitrate, the static LAM should remain the preferred screening model. It is accurate, simple and gains little from time variation. For zinc, the dynamic model is better for prediction. Orthophosphate can use a dynamic interpretation at sites where the diagnostics are acceptable. For ammonia, the results should be presented as evidence of an unresolved change in the data.

For every determinand, source shares should be reported at clearly stated flows and with 95% credible intervals. A share at a common reference flow is useful for comparing sites.

## 6 Conclusions

This study built a censoring-aware workflow for daily river-water concentration data. Repeated samples from the same day were combined so that intensively sampled days did not receive too much weight. Exact, left-censored and interval-censored information was kept. The same chronological split was used for the GAMs, classical LAM and Bayesian models.

The classical LAM was a strong benchmark, especially for nitrate. Its average calibration was better than that of the GAMs. However, some parameter estimates were close to their limits, which showed that the two source components were often less certain than the prediction of total concentration. The hierarchical Bayesian model helped by sharing information across sites and by giving posterior uncertainty.

Allowing the point and diffuse source sizes to change linearly through time helped only some determinands. It gave a clear improvement for zinc and a smaller improvement for orthophosphate. Nitrate was already well described by a static model, and ammonia became worse. The study therefore does not support one dynamic model for every determinand. Model complexity should be chosen from test performance and diagnostics.

Overall, concentration and flow data can help identify pollution patterns, but they cannot identify pollution sources with complete certainty. Source shares should therefore be reported with uncertainty and discussed alongside changes in flow, sampling and laboratory methods. With these limits made clear, the framework provides a practical way to use routine monitoring data to study how pollution patterns differ between sites and change over time.

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